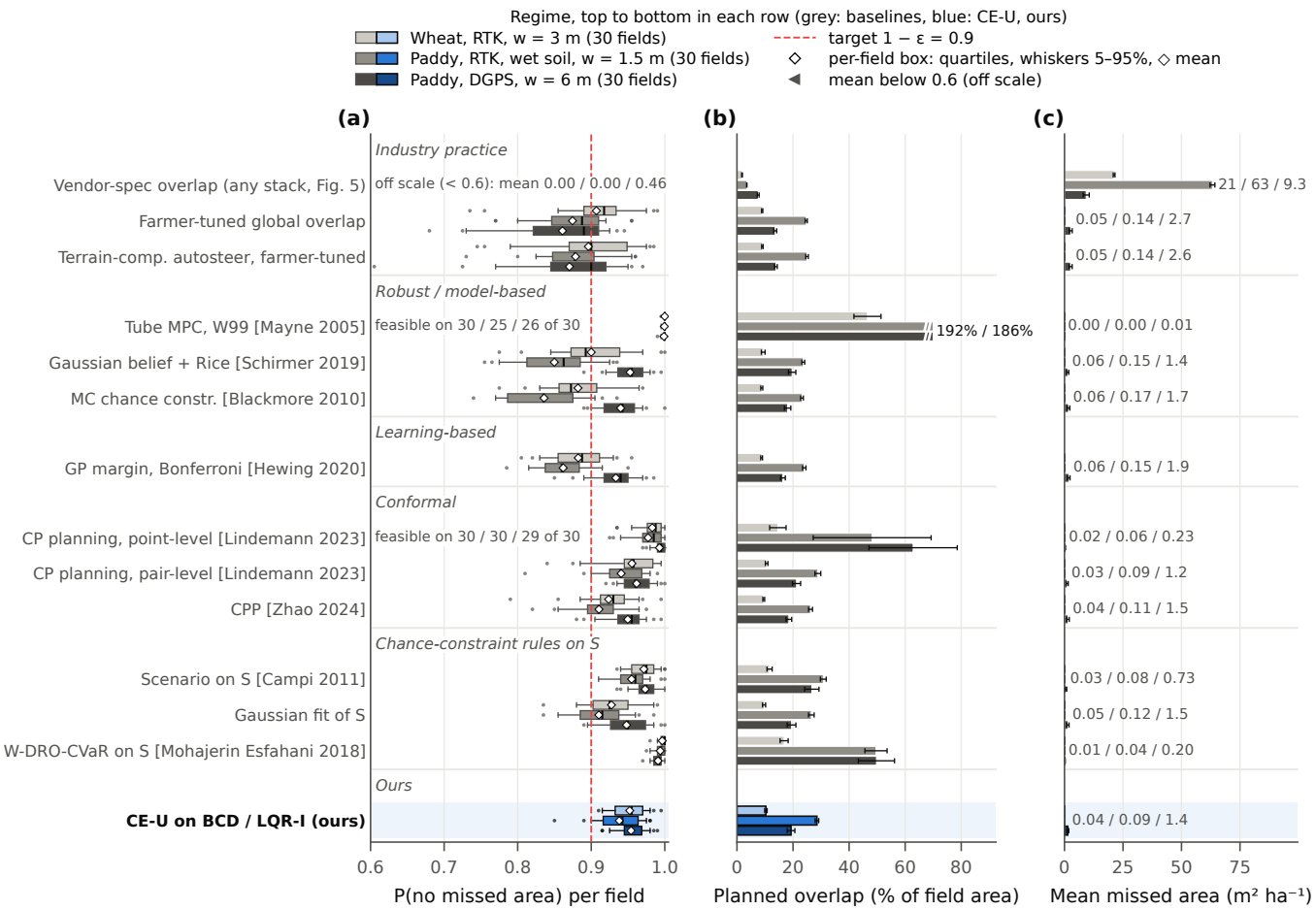

Regime, top to bottom in each row (grey: baselines, blue: CE-U, ours)
Wheat, RTK, w = 3 m (30 fields)
Paddy, RTK, wet soil, w = 1.5 m (30 fields)
Paddy, DGPS, w = 6 m (30 fields)
target 1 − ε = 0.9
per-field box: quartiles, whiskers 5–95%, ◇ mean
mean below 0.6 (off scale)
(a)
(b)
(c)
Industry practice
Vendor-spec overlap (any stack, Fig. 5)
off scale (< 0.6): mean 0.00 / 0.00 / 0.46
Farmer-tuned global overlap
Terrain-comp. autosteer, farmer-tuned
Robust / model-based
Tube MPC, W99 [Mayne 2005]
feasible on 30 / 25 / 26 of 30
Gaussian belief + Rice [Schirmer 2019]
MC chance constr. [Blackmore 2010]
Learning-based
GP margin, Bonferroni [Hewing 2020]
Conformal
CP planning, point-level [Lindemann 2023]
feasible on 30 / 30 / 29 of 30
CP planning, pair-level [Lindemann 2023]
CPP [Zhao 2024]
Chance-constraint rules on S
Scenario on S [Campi 2011]
Gaussian fit of S
W-DRO-CVaR on S [Mohajerin Esfahani 2018]
Ours
CE-U on BCD / LQR-I (ours)
192% / 186%
21 / 63 / 9.3
0.05 / 0.14 / 2.7
0.05 / 0.14 / 2.6
0.00 / 0.00 / 0.01
0.06 / 0.15 / 1.4
0.06 / 0.17 / 1.7
0.06 / 0.15 / 1.9
0.02 / 0.06 / 0.23
0.03 / 0.09 / 1.2
0.04 / 0.11 / 1.5
0.03 / 0.08 / 0.73
0.05 / 0.12 / 1.5
0.01 / 0.04 / 0.20
0.04 / 0.09 / 1.4
0.6
0.7
0.8
0.9
1.0
P(no missed area) per field
0
20
40
60
80
Planned overlap (% of field area)
0
25
50
75
Mean missed area (m² ha⁻¹)